

INTRODUCTION TO NETWORKS & DATA COMMUNICATION

Credit : (02) / Week

Data Communication & Networking

NETWORKS:

- ✓ a group or system of interconnected people or things.
- ✓ interact with others to exchange information and develop professional or social contacts.
- ✓ an arrangement of intersecting horizontal and vertical lines.

DATA COMMUNICATION:

- ✓ refers to the exchange of data between a source and a receiver

WHAT IS A NETWORK?

- ✓ A **network** is a collection of computers, servers, mainframes, **network** devices, peripherals, or other devices connected to one another to allow the sharing of data. An excellent example of a **network** is the Internet, which connects millions of people all over the world
- ✓ A *network* is simply two or more computers that are linked together.
- ✓ The most common types of networks are:
 - *Local Area Networks (LANs)* and
 - *Wide Area Networks (WANs)*.
- ✓ The primary difference between the two is that a *LAN* is generally confined to a limited geographical area, whereas a *WAN* covers a large geographical area. Most *WANs* are made up of several connected *LANs*.

BENEFITS OF A NETWORK

The Uses of a Network

A network connects computers so they can communicate, exchange information, and share resources.

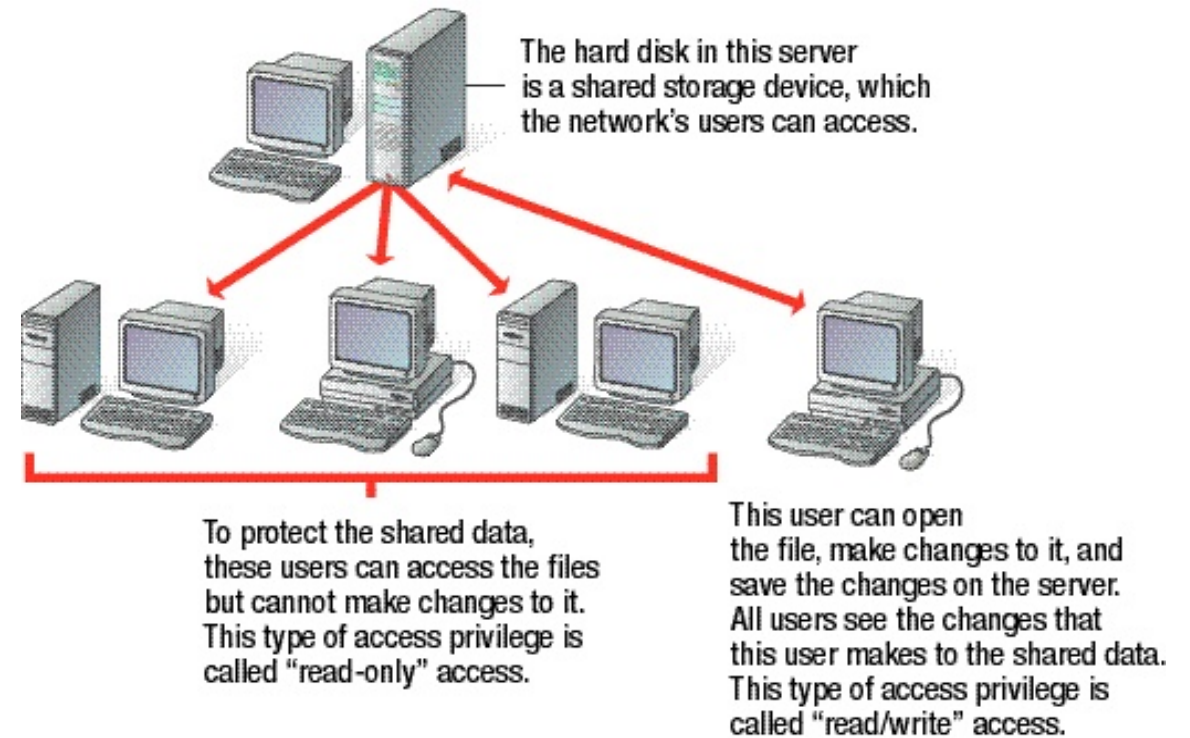
The main benefits of using a network are:

- **Simultaneous Access**
- **Shared Peripheral Devices**
- **Personal Communication**
- **Easier Backup**

BENEFITS OF A NETWORK

The Uses of a Network - Simultaneous Access

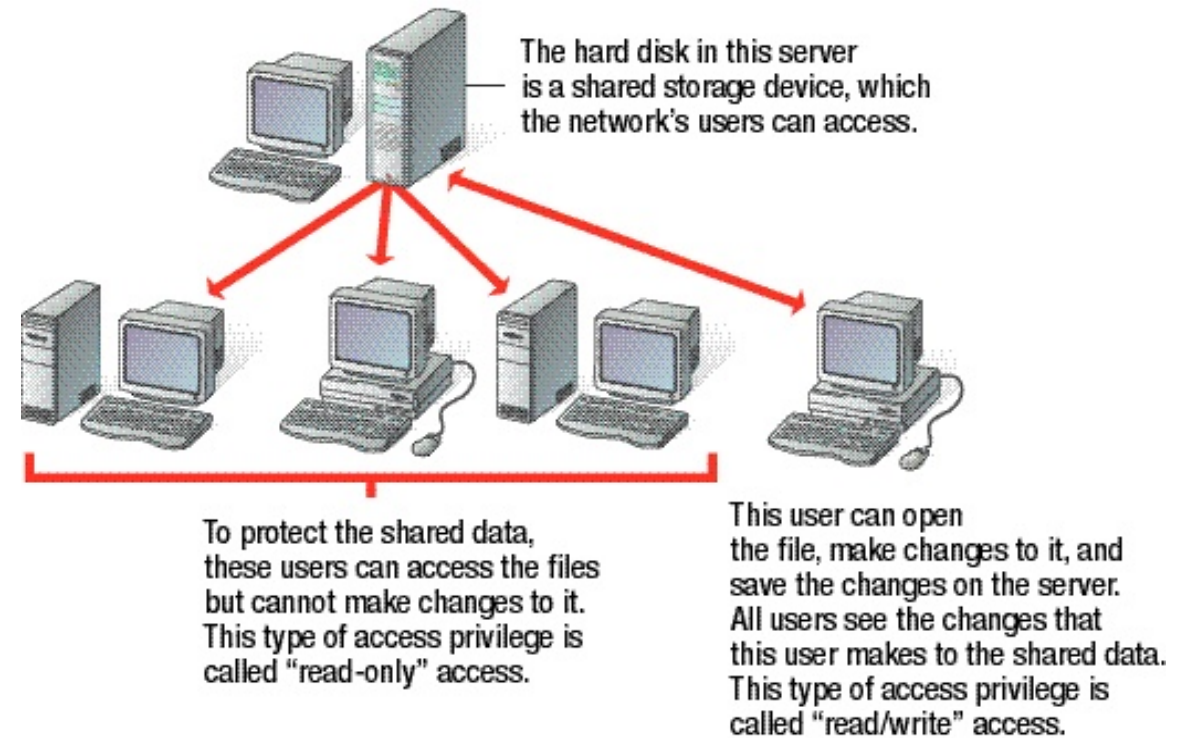
- In organizations, many people may need to use the same data or programs. A network solves this problem.
- Shared data and programs can be stored on a central network server. A server that stores data files may be called a file server.
- Managers may assign access rights to users. Some users may only be able to read data, others may be able to make changes to existing files.



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BENEFITS OF A NETWORK

The Uses of a Network - Shared Peripheral Devices

- Because peripheral (external) devices like printers can be expensive, it is cost-effective to connect a device to a network so users can share it.
- Through a process called spooling, users can send multiple documents (called print jobs) to a networked printer at the same time. The documents are temporarily stored on the server and printed in turn.

The Uses of a Network - Personal Communication

- One of the most common uses of networks is for electronic mail (e-mail).
- An e-mail system enables users to exchange written messages (often with data files attached) across the local network or over the Internet.
- Two other popular network-based communications systems are teleconferencing and videoconferencing.

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BENEFITS OF A NETWORK

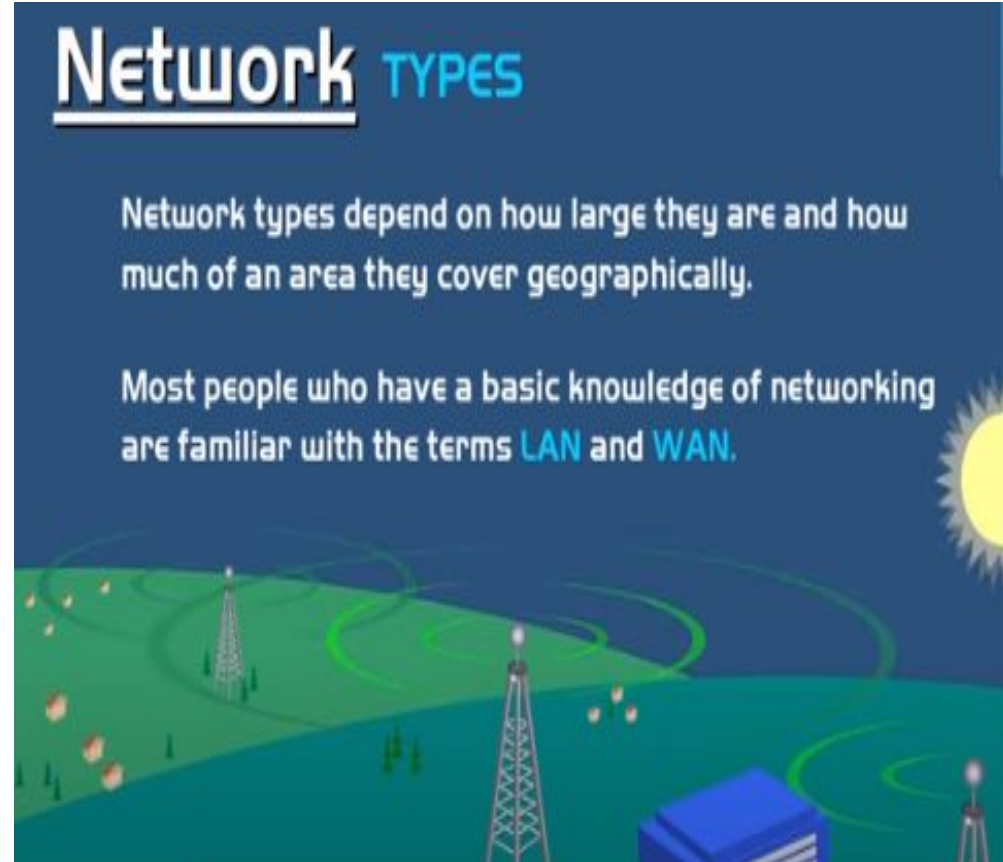
- **Networks enable managers to easily back up (make backup copies of) important data.**
- **Administrators commonly back up shared data files stored on the server, but may also use the network to back up files on users' PCs**
- **Information sharing:** Authorized users can use other computers on the network to access and share information and data. This could include special group projects, databases, etc.
- **Hardware sharing:** One device connected to a network, such as a printer or scanner, can be shared by many users.
- **Software sharing:** Instead of purchasing and installing a software program on each computer, it can be installed on the server. All of the users can then access the program from a single location.
- **Collaborative environment:** Users can work together on group projects by combining the power and capabilities of diverse equipment.

RISKS OF NETWORK COMPUTING

- The security of a computer network is challenged everyday by:
 - Equipment malfunctions
 - System failures
- Note: equipment malfunctions and system failures may be caused by natural disasters such as floods, storms, or fires, and electrical disturbances
 - Computer hackers
 - Virus attacks

TYPES OF NETWORKS

- *Local Area Network (LAN)* –
- *PAN*
- *CAN*
- *MAN*
- *Wide Area Network (WAN)*



TYPES OF NETWORKS

Personal Area Network, or PAN

- ✓ A **personal area network**, or **PAN**, is a **computer network** that enables communication between **computer** devices near a person.
- ✓ **Examples:** PANs can be wired, such as USB or FireWire, or they can be wireless, such as infrared, Bluetooth and ultra wideband, or UWB.
- ✓ The range of a **PAN** typically is a few meters.

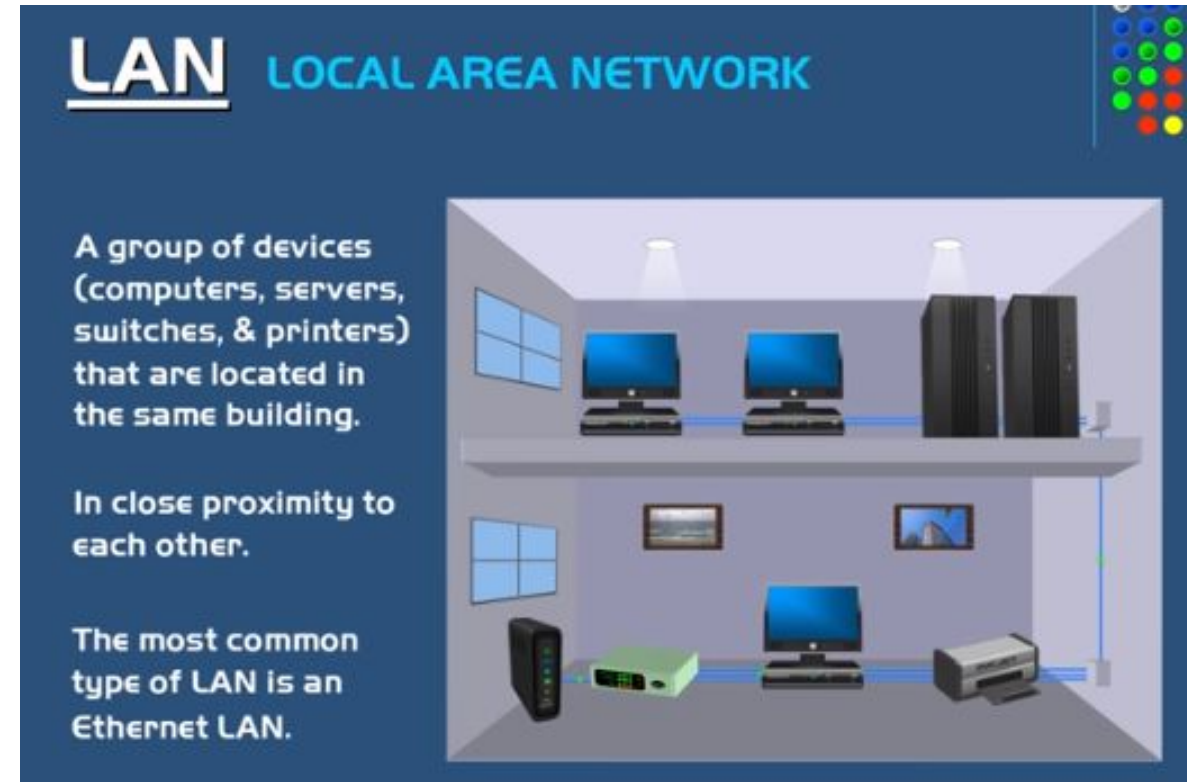


TYPES OF NETWORKS

Local Area Network (LAN) –

- a network that spans a small area
- such as a building or an office.

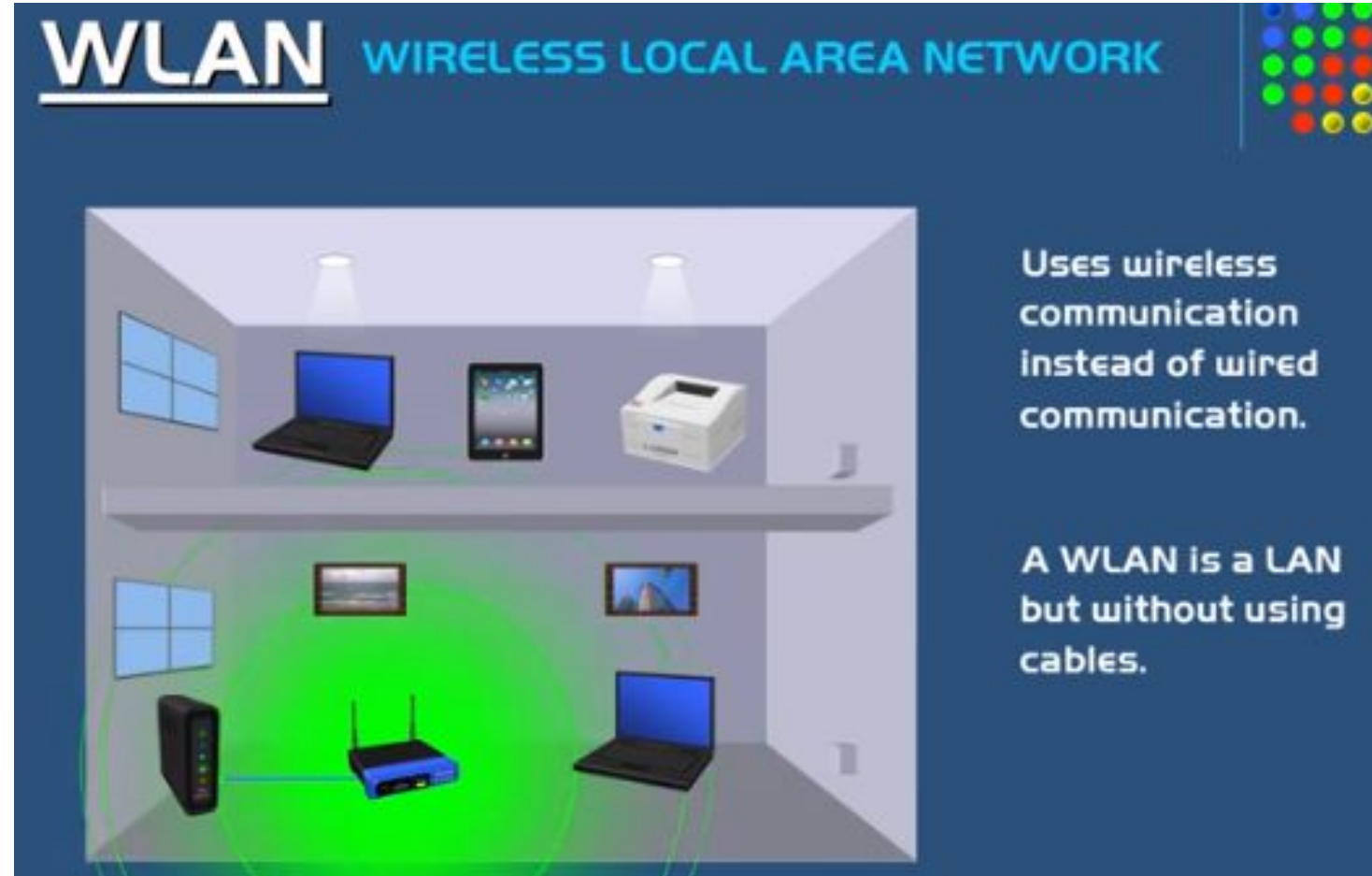
- Software applications and other resources are stored on a *file server*.
- *Print servers enable* multiple users to share the same printer.



TYPES OF NETWORKS

• *Local Area Network (LAN) – Wireless Local Area Network (WLAN):*

- ✓ WLANs use radio, infrared and microwave transmission to transmit data from one point to another without cables.
- ✓ Therefore WLAN offers way to build a Local Area Network without cables.



• **Advantages of LAN**

Given below are the various advantages of LAN:

- ✓ In an office which is connected via LAN network, we can share the hardware and software resources like printers, FAX, drivers and hard-disk as they are on one platform and thus this type of network turns out to be cost-effective.
- ✓ As being connected on the network, the offices or firms using the same type of software for job purposes need not purchase separately for each of the host clients as the software can easily be shared with everyone on an equal level.
- ✓ LAN network works as a client-server model, therefore data can be stored centrally on one PC called as a server in a network and it can be accessible to all the other client PC's via LAN. By following this method, we need not store data locally at one single node.
- ✓ Communication will be handy and economical by using LAN network.

• **The disadvantages of LAN are:**

- ✓ LAN networks come out to be cost-effective and time-saving, as we can share various resources on one platform. However, the initial installation cost of the network is very high.
- ✓ It is having a geographical area limitation and can only cover a small area (1-5 km).
- ✓ As it works on a single cable, if it gets faulty then the overall network will stop working. Hence, it needs a full -time maintenance officer called an administrator.
- ✓ Crucial Data of offices or factories is saved on a single server which can easily be accessible by all the nodes thus it is having all time data security issues as any unauthorized person can also access the confidential data.

TYPES OF NETWORKS

Campus Area Network (CAN)

- ✓ A CAN is a network that covers an educational or corporate campus.
- ✓ A campus area network is larger than a local area network **LAN** since it may span multiple buildings within a specific area.

Examples include elementary schools, university campuses, and corporate buildings.



TYPES OF NETWORKS

Metropolitan Area Network (MAN)

- ✓ MAN covers a bigger geographical area than LAN Network
- ✓ E.g. cities and districts.
- ✓ It can also be considered as a superior version of the LAN network.,
- ✓ As LAN covers only a minor area of the network,
- ✓ MAN is designed to connect a city or two villages together through it.
- ✓ The area covered by MAN is generally 50-60 km.
- ✓ Fiber optical cable and twisted pair cables are used for connectivity for communication through MAN networks.
- ✓ MAN can also be considered as a group of one or more LAN networks connected together through a single cable.
- ✓ ATM are the common protocol practice for communication in MAN



• Application of MAN

- For Example, MAN can be used to connect various police stations which are situated within a district or city with each other. The officers can easily communicate with each other and quickly pass on the important data and an urgent message over this network without any need for the Internet connection.
- #2) Any private firm can also use a MAN network for inter-connectivity between their offices situated at two different towns of a district. The firm can share resources like data file, images, software & hardware parts etc., with each other. Thus it provides resource sharing over a large distance than the LAN networks

TYPES OF NETWORKS

• Advantages of MAN

- **Given below are the various advantages of MAN:**

- ✓ It is very efficient and swift for communication over fiber optic cable for interconnection of networks in cities.
- ✓ It serves many villages and cities and thus provides great inter-connectivity at a low cost.
- ✓ It works on ring or bus topology with a protection link, thus data can be transmitted or received simultaneously over nodes and if one link fails the other will keep the network live.

• Disadvantages of MAN

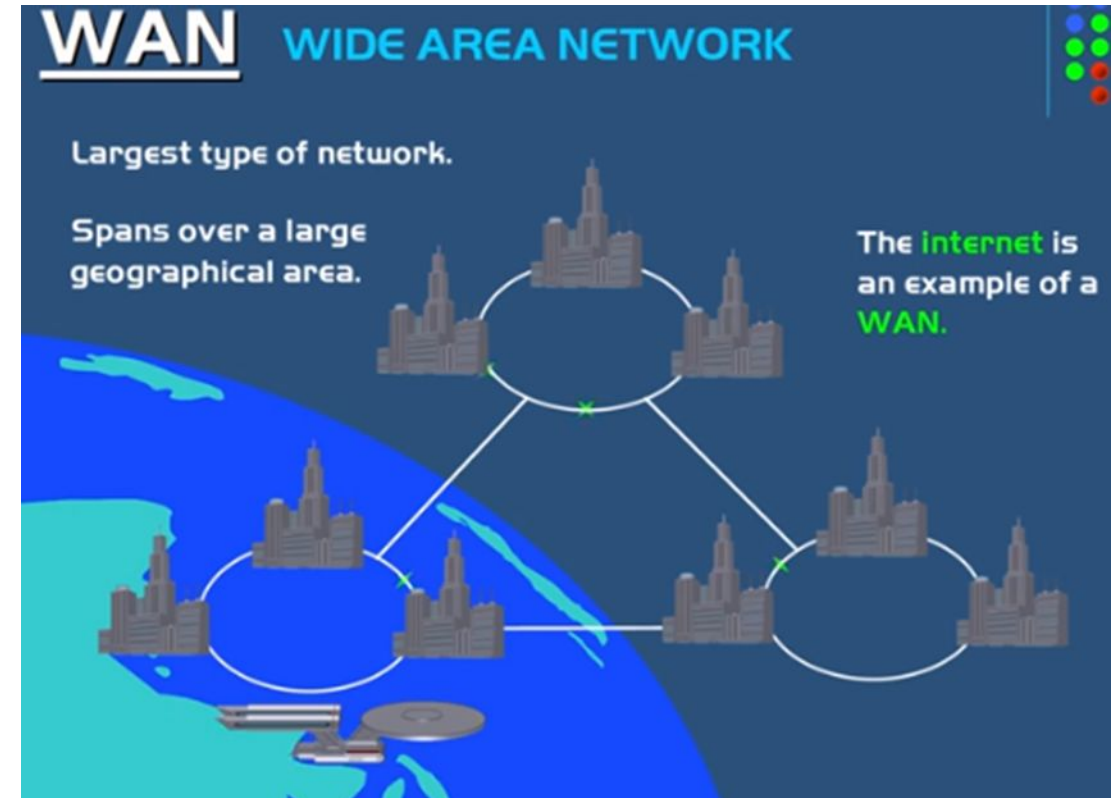
- **The disadvantages of MAN are:**

- ✓ Depending upon the distance between two nodes, the cable length required for inter-connection differs every time. Thus greater will be the cable length, the more will be the cost of the network.
- ✓ Security is a big concern for this network as for such a big distance anyone can hack the network. We can't put security at each level of the network, hence it becomes easier for unwanted people to access it for their own benefits.

TYPES OF NETWORKS

Wide Area Network (WAN) –

- a network that spans a
- wide geographical area;
- there are several types of WANS
- WAN consist of
 - Metropolitan area network (MAN)
 - CAN (Campus Area Network)
 - Public access network (PAN)



COMMUNICATIONS MEDIA

• COMMUNICATIONS CHANNEL:

- To transfer data from one computer to another requires some type of link through which the data can be transmitted. This link is known as the *communications channel*.
- To send data through the channel requires some type of [*transmission media*](#), which may be either physical or wireless

(a) PHYSICAL MEDIA

- *Twisted-pair cable* – consists of two independently insulated wires twisted around each other (least expensive type of cable—the kind that is used in many telephone systems)
- *Coaxial cable* – consists of an insulated center wire grounded by a shield of braided wire (the primary type of cabling used by the cable television industry; it is more expensive than twisted pair)
- *Fiber-optic cable* – contains hundreds of clear fiberglass or plastic fibers (threads) (made from thin, flexible glass tubing; bandwidth is greater, so it can carry more data; it is lighter than metal wires and is less susceptible to interference; it is fragile and expensive)
- *ISDN line* – a special digital telephone line that transmits and receives information at very high speeds

• (b) WIRELESS MEDIA

- *Microwave system* – transmits data via high-frequency radio signals through the atmosphere
- *Satellite system* – receive transmitted signals, amplify them, and then transmit the signals to the appropriate locations
- *Cellular technology* – uses antennae resembling telephone towers to pick up radio signals within a specific area (cell)
- *Infrared technology* – transmits data as infrared light waves from one device to another, providing wireless links between PCs and peripherals

• **Note:** *The type selected is determined by the type of network, the size of the network, and the cost*

NETWORKING HARDWARE AND SOFTWARE

- **Hub** – electronic device (with a number of ports) used in a LAN to link groups of computers
- **Switch** A network switch is a computer networking device that is used to connect many devices together on a computer network. A switch is considered more advanced than a hub because a switch will only send message to device that needs or request it.
- **Routers** - electronic devices used to ensure messages are sent to their intended destinations
- **Gateway** – consists of hardware and/or software that allows communications between dissimilar networks
- **Bridge** – consists of hardware and/or software that allows communication between two similar networks

Hub, Switch, & Router



HUB

Only detects that a device is physically connected to it.



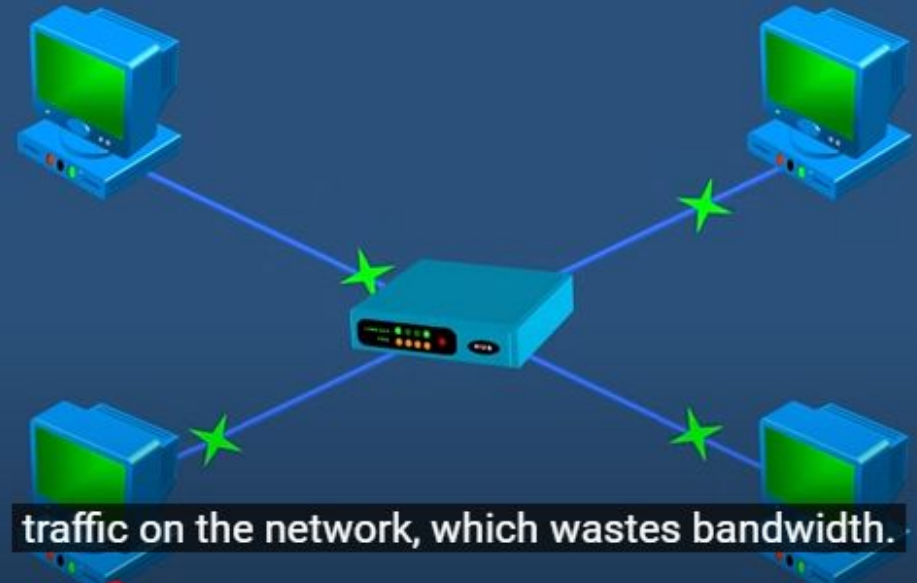
SWITCH

Can detect specific devices that are connected to it.

Keeps a record of the MAC addresses of those devices.

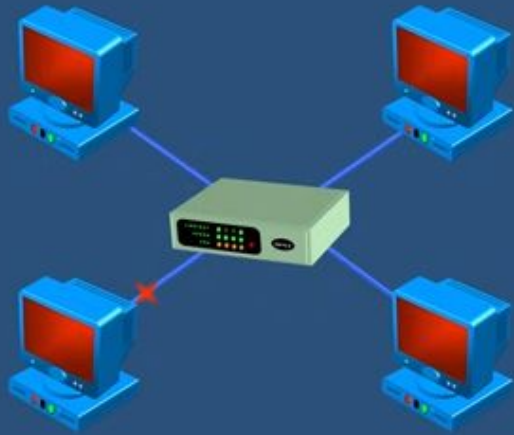
that are connected to it because it keeps a record of the MAC addresses of those devices.

Hub





Hub, Switch, & Router



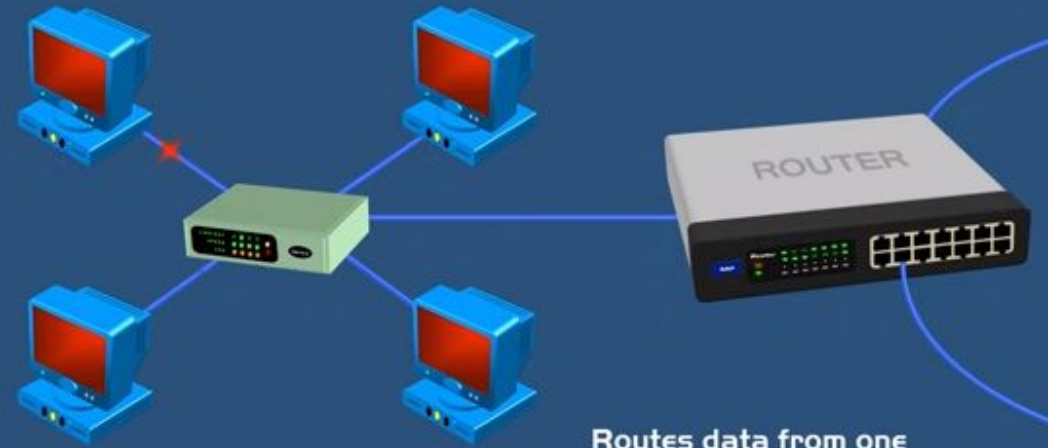
Hubs and Switches are used to exchange data within a local area network.

Not used to exchange data outside their own network.

To exchange data outside their own network, a device needs to be able to read I.P. addresses.

addresses. And hubs and switches do not read IP addresses. So that's where the router comes in.

Router



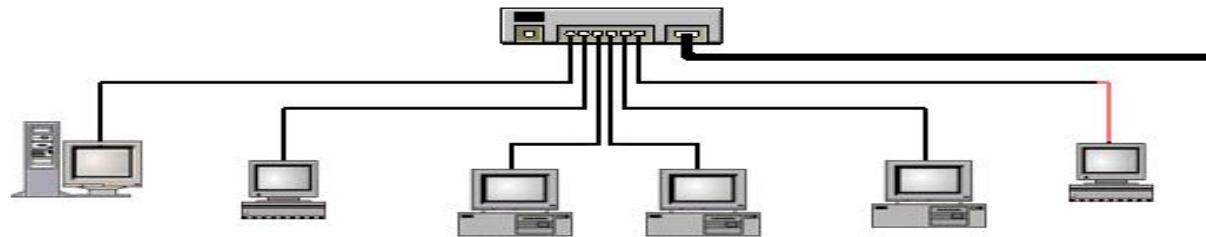
Routes data from one network to another based on their IP address.

from one network to another

NETWORKING HARDWARE AND SOFTWARE

HUB:

- Simply put, the hub's major function is to replicate data it receives from one device attached to it to all others.



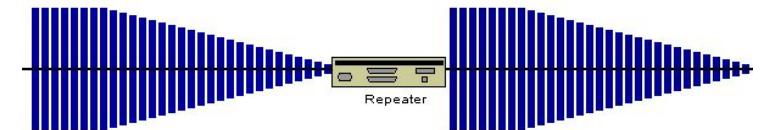
REPEATERS

- ✓ A common problem in the networking world is that of weakening electrical signals.
- ✓ A repeater is connected to two cable segments. Any electrical signal reaching the repeater from one segment, will be amplified and retransmitted to the other segment.

An illustration of a repeater at work

The electrical signal entering the repeater at one end is weakened.

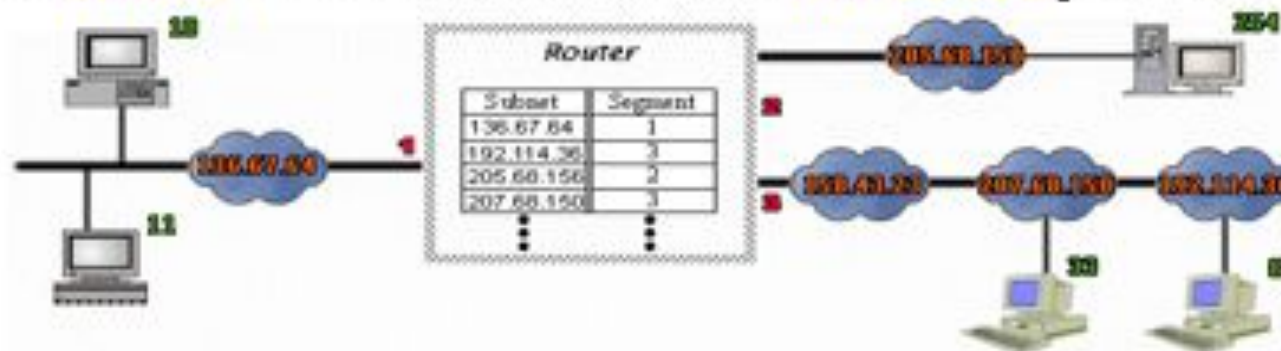
The repeater amplifies the electrical signals and resends the data



NETWORKING HARDWARE AND SOFTWARE

ROUTERS

- ✓ Routers **connect** two or more networks and **forward data** packets between them.



GATEWAY:

- ✓ The term gateway now refers to special-purpose devices, that perform **protocol conversions**. Gateways implement application layer conversions of information received from various protocols therefore help different network to communicate.



HOW ARE NETWORKS CATEGORIZED?

Networks are usually classified using three properties:

- *Topology*
- *Protocol*
- *Architecture*

TYPES NETWORK TOPOLOGIES

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection

- *Network Topology (or layout)* – refers to the way computers and peripherals are configured to form networks
 - *Bus topology* – all computers are linked by a single line of cable
 - *Star topology* – multiple computers and peripheral devices are linked to a central computer, called a *host*
 - *Ring topology* – each computer or workstation is connected to two other computers, with the entire network forming a circle
 - *Hybrid topology* – combine network layout types to meet their needs

NETWORK TOPOLOGIES

BUS Topology

Bus topology is a network type in which every computer and network device is connected to single cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.



Features of Bus Topology

1. It transmits data only in one direction.
2. Every device is connected to a single cable

Advantages of Bus Topology

1. It is cost effective.
2. Cable required is least compared to other network topology.
3. Used in small networks.
4. It is easy to understand.

5. Easy to expand joining two cables together.

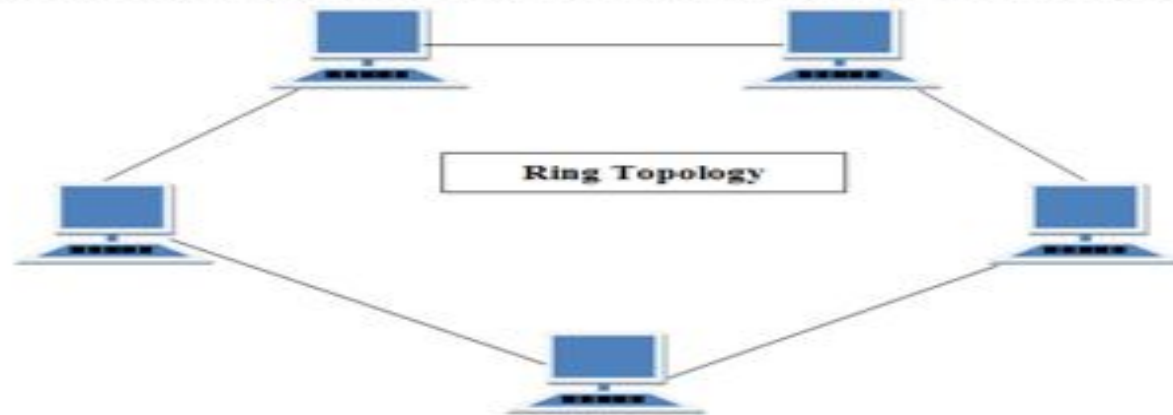
Disadvantages of Bus Topology

1. Cables fails then whole network fails.
2. If network traffic is heavy or nodes are more the performance of the network decreases.
3. Cable has a limited length.
4. It is slower than the ring topology

NETWORK TOPOLOGIES

RING Topology

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbours for each device.



Features of Ring Topology

1. A number of repeaters are used for Ring topology with large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
2. The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called **Dual Ring Topology**.
3. In Dual Ring Topology, two ring networks are formed, and data flow is in opposite direction in them. Also, if one ring fails, the second ring can act as a backup, to keep the network up.
4. Data is transferred in a sequential manner that is bit by bit. Data transmitted, has to pass through each node of the network, till the destination node.

Advantages of Ring Topology

1. Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
2. Cheap to install and expand

Disadvantages of Ring Topology

1. Troubleshooting is difficult in ring topology.
2. Adding or deleting the computers disturbs the network activity.
3. Failure of one computer disturbs the whole network.

NETWORK TOPOLOGIES

•STAR Topology

•In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node.

•Features of Star Topology

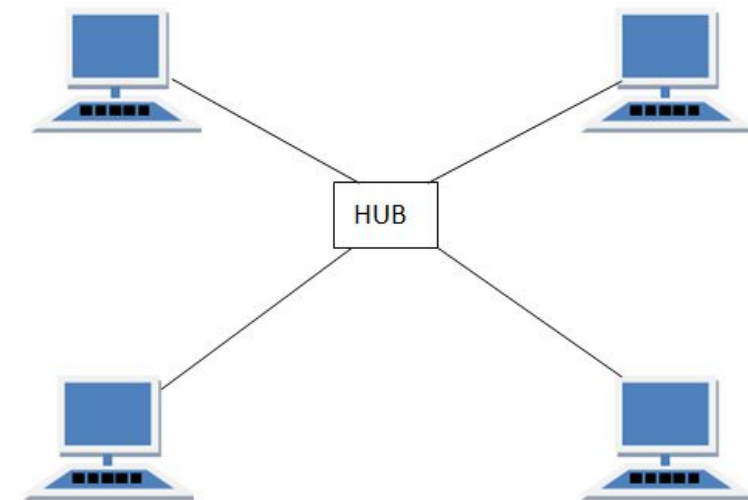
- Every node has its own dedicated connection to the hub.
- Hub acts as a repeater for data flow.
- Can be used with twisted pair, Optical Fibre or coaxial cable.

•Advantages of Star Topology

- Fast performance with few nodes and low network traffic.
- Hub can be upgraded easily.
- Easy to troubleshoot.
- Easy to setup and modify.
- Only that node is affected which has failed, rest of the nodes can work smoothly.

•Disadvantages of Star Topology

- Cost of installation is high.
- Expensive to use.
- If the hub fails then the whole network is stopped because all the nodes depend on the hub.
- Performance is based on the hub that is it depends on its capacity



NETWORK TOPOLOGIES

•MESH Topology

- It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other.

Mesh has $n(n-1)/2$ physical channels to link n devices.

- There are two techniques to transmit data over the Mesh topology, they are :

- Routing
- Flooding

•MESH Topology: Routing

- In routing, the nodes have a routing logic, as per the network requirements. Like routing logic to direct the data to reach the destination using the shortest distance. Or, routing logic which has information about the broken links, and it avoids those node etc. We can even have routing logic, to re-configure the failed nodes.

•MESH Topology: Flooding

- In flooding, the same data is transmitted to all the network nodes, hence no routing logic is required. The network is robust, and the its very unlikely to lose the data. But it leads to unwanted load over the network.

NETWORK TOPOLOGIES

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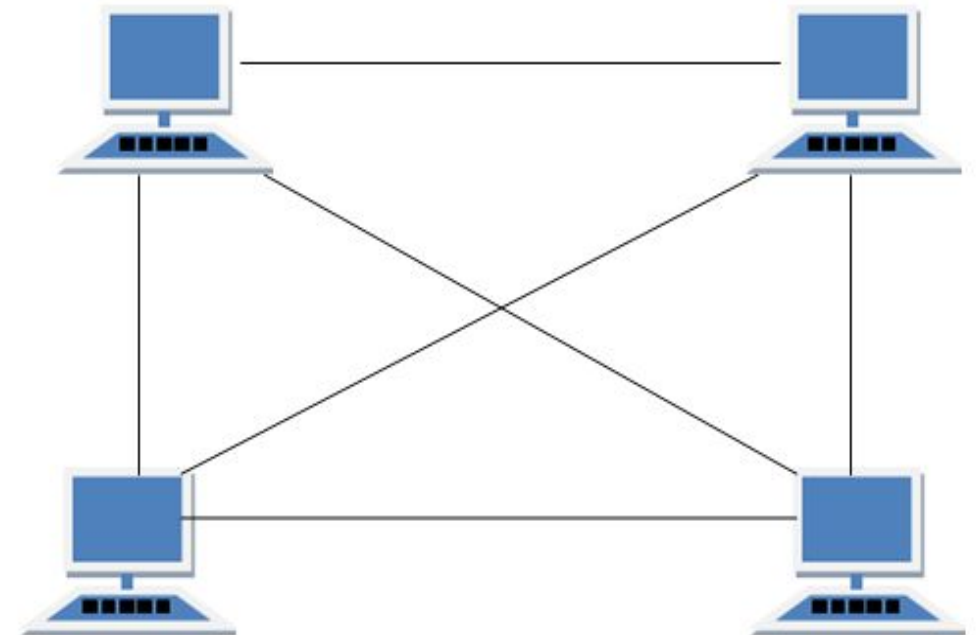
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•In flooding, the same data is transmitted to all the network nodes, hence no routing logic is required. The network is robust, and the its very unlikely to lose the data. But it leads to unwanted load over the network.

$$N=4 \quad \square \quad n(n-1)/2 \quad \square \quad 4(4-1)/2$$

$$12/2 = 6$$

Physical connections are required



NETWORK TOPOLOGIES

Types of Mesh Topology

Partial Mesh Topology : In this topology some of the systems are connected in the same fashion as mesh topology but some devices are only connected to two or three devices.

Full Mesh Topology : Each and every nodes or devices are connected to each other.

Features of Mesh Topology

Fully connected.

Robust.

Not flexible.

Advantages of Mesh Topology

Each connection can carry its own data load.

It is robust.

Fault is diagnosed easily.

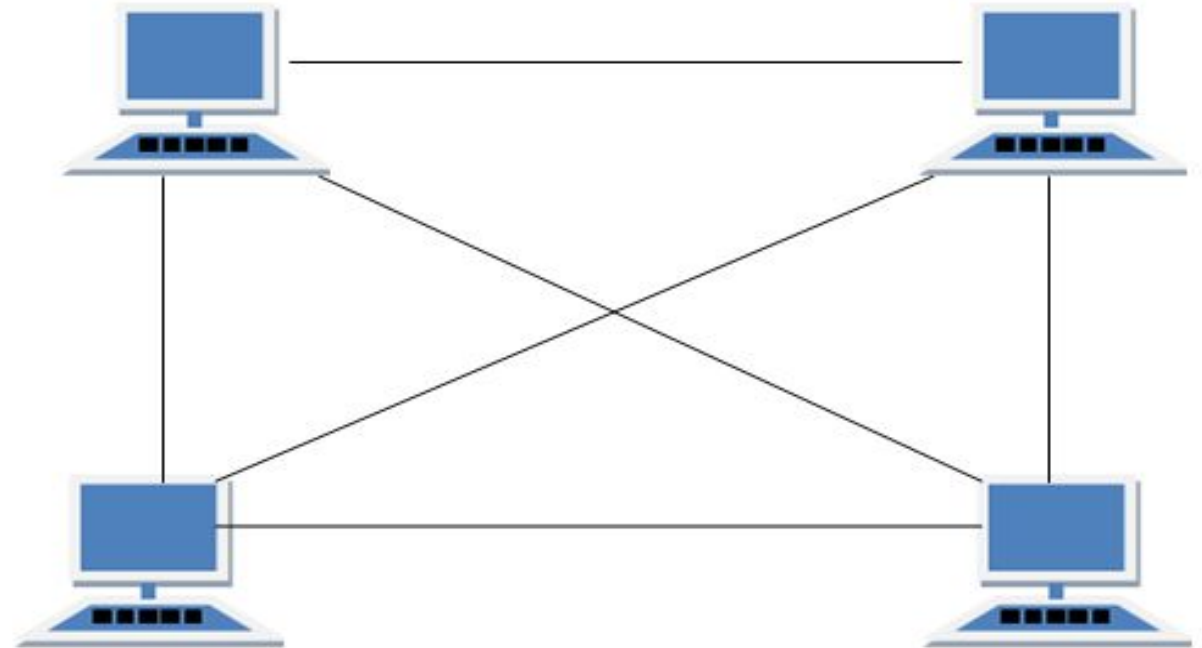
Provides security and privacy.

Disadvantages of Mesh Topology

Installation and configuration is difficult.

Cabling cost is more.

Bulk wiring is required.



NETWORK TOPOLOGIES

TREE Topology

- It has a root node and all other nodes are connected to it forming a hierarchy. It is also called hierarchical topology. It should at least have three levels to the hierarchy.

•Features of Tree Topology

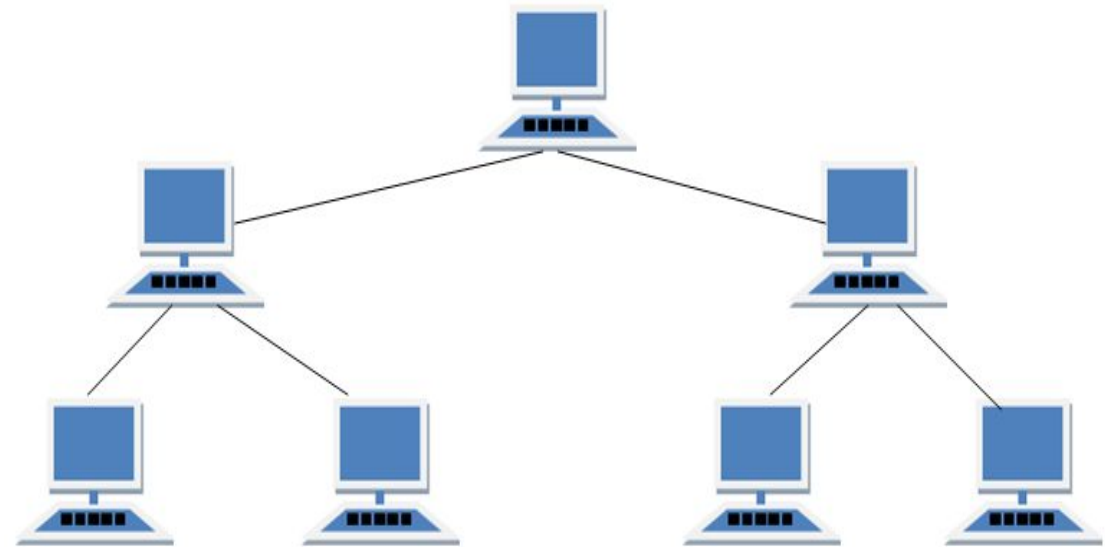
- Ideal if workstations are located in groups.
- Used in Wide Area Network.

•Advantages of Tree Topology

- Extension of bus and star topologies.
- Expansion of nodes is possible and easy.
- Easily managed and maintained.
- Error detection is easily done.

•Disadvantages of Tree Topology

- Heavily cabled.
- Costly.
- If more nodes are added maintenance is difficult.
- Central hub fails, network fails.



NETWORK TOPOLOGIES

•HYBRID Topology

- It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

• Features of Hybrid Topology

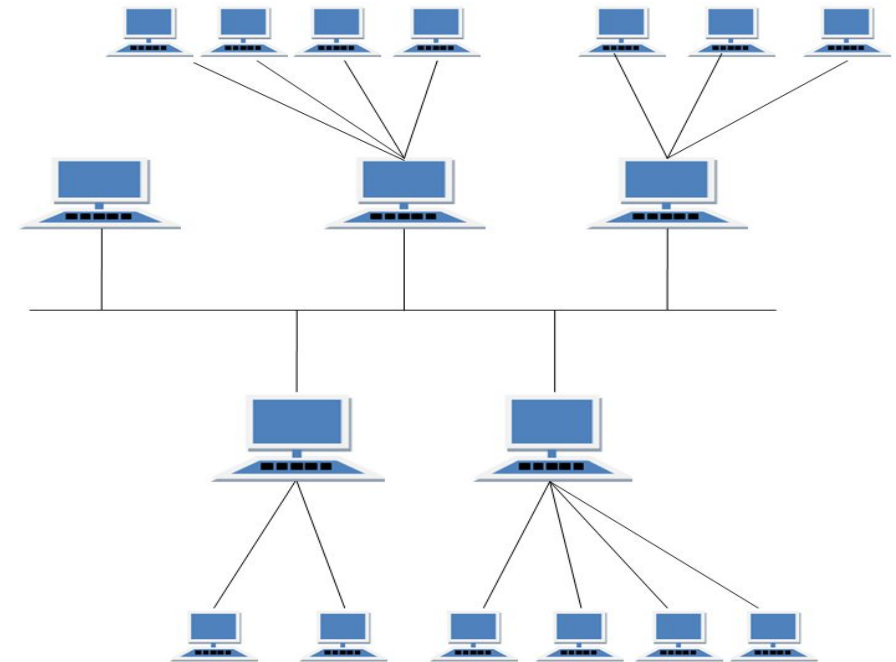
- It is a combination of two or topologies
- Inherits the advantages and disadvantages of the topologies
- included

•Advantages of Hybrid Topology

- Reliable as Error detecting and trouble shooting is easy.
- Effective.
- Scalable as size can be increased easily.
- Flexible.

•Disadvantages of Hybrid Topology

- Complex in design.
- Costly.



NETWORK/COMMUNICATION PROTOCOLS

- **NETWORK/COMMUNICATION PROTOCOLS**
- **A protocol** is simply an agreed-on set of rules and procedures for transmitting data between two or more devices.
- **Features determined by the protocol are:**
 - How the sending device indicates it has finished sending the message.
 - How the receiving device indicates it has received the message.

NETWORK/COMMUNICATION PROTOCOLS

•COMMUNICATION PROTOCOLS

- ✓ *File transfer protocol (FTP)* – used to transmit files.
- ✓ *Simple mail transfer protocol (SMTP)* – used to transmit e-mail messages.
- ✓ *Post office protocol (POP)* – allows the recipient to retrieve messages.
- ✓ *Wireless application protocol* – enables wireless devices to access and use the Internet using a client/server network.
- ✓ *802.11 – protocol* for wireless LAN technology

NETWORK ARCHITECTURE

- *Network architecture* – refers to the way a network is designed and built. The two major types are:
 - *Peer-to-peer architecture* – Computers connect with each other in a workgroup to share files, printers, and Internet access. This is used to connect a small number of computers.
 - *Client/server architecture* – sends information from a client computer to server, which then relays the information back to client computer, or to other computers on the network

- **DATA TRANSMISSION CHARACTERISTICS**

- ☐ ***Bandwidth*** (rate of transmission)
- ☐ ***Analog or Digital*** (type of signal)
 - ✓ *Analog transmission* – takes the form of continuous waves transmitted over a medium at a certain frequency range
 - ✓ *Digital transmission* – sends data in the form of bits
- ☐ ***Serial or Parallel*** (order of bits)
 - ✓ *Serial transmission* – all of the data bits are transmitted one bit after another in a continuous line
 - ✓ *Parallel transmission* – data bits are sent at the same time along multiple paths

- **COMMUNICATIONS SOFTWARE**

- ☐ ***E-Mail Software*** – used to send and receive electronic messages using the Internet
- ☐ ***Web Browser*** – allows users to access and view Web pages
- ☐ ***Groupware*** – allows groups of people on a network to share information and to collaborate on various projects

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END OF LECTURE

Any Questions !!!